How much vibe is the coding?

My journey from using AI on my projects

Agenda

- Context - What are we talking about?
- Overview of the project DNSServer.DebugLogParser
- How it was done

Get-Person -Short

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IT infrastructure guy and PowerShell fellow

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Security

Context

Context - What are we talking about?

- DDI topics – Moving DNS
  - Network teams usually are not in love Windows
- Considerations needs to be made!
  - Challenges on moving DNS
  - Possibilities to gain control
- Why not a script, but a module?
  - Isn't it just a one-liner?
- Yet another solution… wasn't it done before?

Actually, it was done before

```
PS C:\Users\andre> Find-PSResource DNSDebugLogHandler | fl

AdditionalMetadata       : {[NormalizedVersion, 1.0.3]}
Author                   : Johannes Schoepp
CompanyName              :
Copyright                : (c) Johannes Schoepp. All rights reserved.
Dependencies             : {}
Description              : This module parses Windows DNS Server Debug logfiles.

                           It is based on the excellent DNSLogModule by virot. It was ported with his permission.
IconUri                  :
Includes                 : Microsoft.PowerShell.PSResourceGet.UtilClasses.ResourceIncludes
InstalledDate            :
InstalledLocation        :
IsPrerelease             : False
LicenseUri               : https://github.com/jschpp/DNSDebugLogHandler/blob/main/LICENSE
Name                     : DNSDebugLogHandler
Prerelease               :
ProjectUri               : https://github.com/jschpp/DNSDebugLogHandler
PublishedDate            : 20.07.2020 10:34:47
ReleaseNotes             : https://github.com/jschpp/DNSDebugLogHandler/blob/main/CHANGELOG.md
Repository               : PSGallery
RepositorySourceLocation : https://www.powershellgallery.com/api/v2
Tags                     : {DNS, parsing, log, logfile…}
Type                     : Module
UpdatedDate              :
Version                  : 1.0.3
```

PSModule DNSDebugLogHandler

- Let's check it out… https://github.com/jschpp/DNSDebugLogHandler

README MIT license

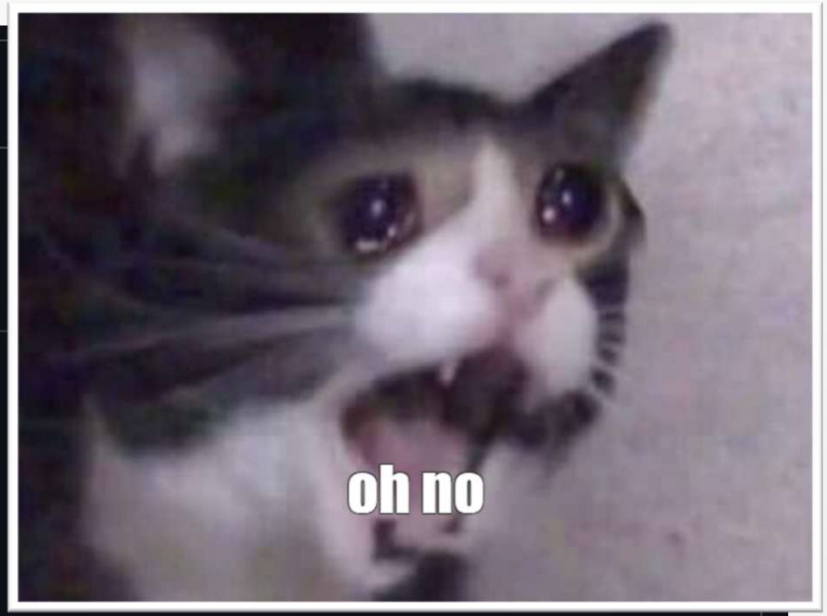
DNS Debug Logfile handler

Azure Pipelines succeeded DNSDebugLogHandler v1.0.3

This module parses dns debug log files.

At this moment it is only build for PowerShell Version 7 and up.

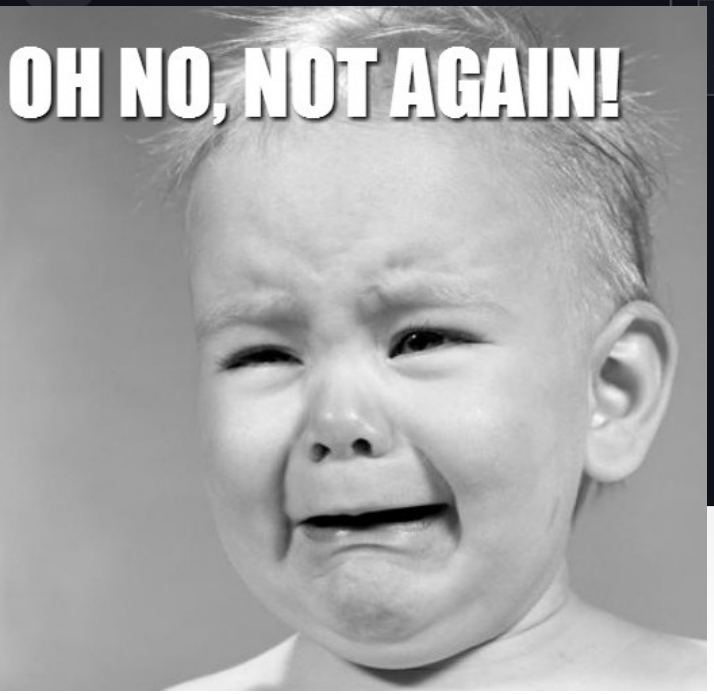
It is based on the excellent DNSLogModule by @virot. It was ported with his permission.



PSModule DNSLogModule

- Let's check it out… again
  - https://github.com/virot/DNSLogModule

Files



DNSLogModule / Includes / Get-DNSDebugLog.ps1

Code Blame 114 lines (101 loc) · 4.06 KB

```
80      }
81      Process
82      {
83        Write-Verbose "Getting the contents of $Path, and matching for correct rows."
84        $rows = ([array](Get-Content $Path)) -notmatch 'ERROR offset' -notmatch 'NOTIMP'
85        Write-Verbose "Found $($rows.count) in debuglog, processing 1 at a time."
86        ForEach ($row in $rows)
87        {
88          write-verbose "Row: $($row)"
89          $match = [regex]::match($row,$dnspattern)
```

I got a feeling…

This will take some more energy

…but maybe I can challenge my GitHub Copilot

Overview and Creative Process

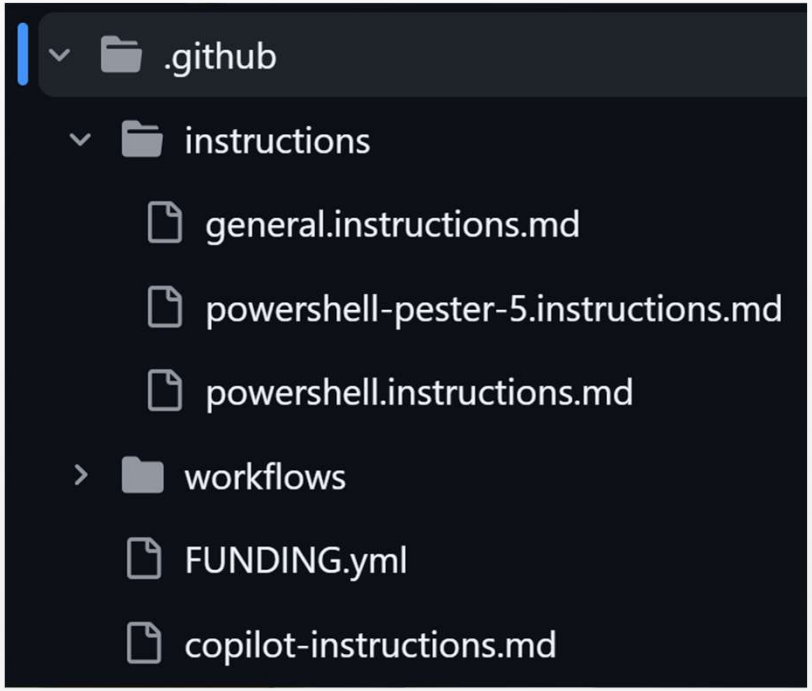
https://github.com/AndiBellstedt/DNSServer.DebugLogParser

Overview to DNSServer.DebugLogParser

- Code base 👉 PSMD template
  - No dependency
  - Well defined folder structure
  - GitHub actions - build stuff (CI/CD) already given
  - Predefined Pester Tests
- Additional stuff
  - Assets
  - Docs
  - Changelog
  - .github folder

The AI stuff

- Markdown everywhere
- Guidelines for the AI
  - copilot-instruction.md
- General project guidelines
  - General.instructions.md
- Language specific instructions
  - Powershell*.instructions.md



From script to full blown module

- First prompt via Copilot was okay
  - Keep in mind!
    AI is doing what you tell it to do
- Various iterations on the script
  - Let it cook, let it cook
  - Perfectionism was in there for sure
- Considerations about preserving and ease of distribution
- Take care about helper functions
- Introduce pester tests
- Write documentation

Demo

(also called... the switch into the interesting part)

Conclusion

- DON'T PUT BLIND TRUST IN AI RESULTS
  - It might be risky
- AI is actually helpful for doing painful work
- The DNSServerDebugLogParser module is a special-purpose tool
  - Engineered as a tool, not a single-purpose static solution
- There is more of the story…
  - Usage within a Windows DNS (Domain Environment)
  - Utilizing the results – Reporting outcomes

Question & Answers

– Thank you –

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